UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 011 (Legendre) · Legendre's and Cauchy's linear differential equations · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011L·8	Unit 2 deck 011 “Legendre's differential equation” (SESSION 9), slide 8 (rendered .pptx)
T011L 00:05:20	AV-summary transcript “Legendre differential equation” (21.9 min, listed first), segment start
T011Lb 00:12:26	AV-summary transcript “Legendre differential equations_Continued” (23 min, second), segment start
CW U2 P30	Unit 1 & 2 class-work sheet, Unit 2 problem 30
ESA Jul-23 3a	Past paper: UE22MA141A July 2023, question 3.a (Jan-24 s1...s4 = UE23MA141A Jan 2024 sets 1–4)

Contents

1. Figure G2.1 · The new variable: $z = \log(ax+b)$ and $z = \log x$
2. Figure G2.2 · Complex roots: $\cos(\log(1+x))$ oscillates ever more slowly
3. Figure G3.1 · Slide example: $y_p = \frac{3}{4} - \frac{3(2x+3)}{16}$ and general solutions
4. Figure G4.1 · ESA Jan-24 s3 3d: $y_p = -x^2 \sin x$ and the CF terms x^2, x^3

The deck has no graphs, so every figure is added for clarity. Colours follow the Okabe–Ito order: blue = the particular integral or the main curve, vermilion (dashed) = the right-hand side or a second curve, green and purple = CF terms; grey for asymptotes, guides and sample general solutions. Every plotted solution was checked by substituting it back with sympy.

2 The substitution $ax + b = e^z$ and the operator results

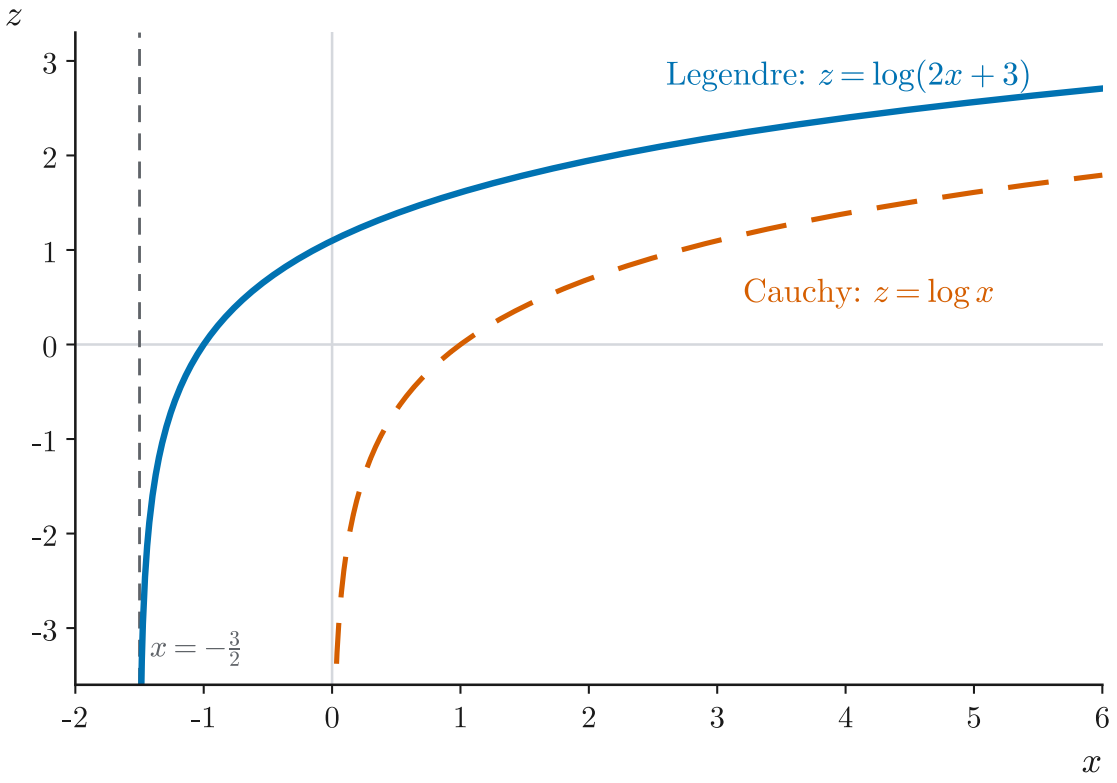


Figure G2.1. The new variable: $z = \log(ax + b)$ and $z = \log x$. ESA ★★

ADDED FOR CLARITY S011L·5 · T011L 00:05:20 · T011L 00:18:30

How to read it. Blue: $z = \log(2x + 3)$ (slide example, $a = 2$, $b = 3$), defined for $x > -\frac{3}{2}$ (grey asymptote). Vermilion dashed: $z = \log x$ (Cauchy), defined for $x > 0$. Both are increasing, so each x has one z and back. Equal steps in z are ever larger steps in x: e^{mz} in z becomes the power $(ax + b)^m$ in x.

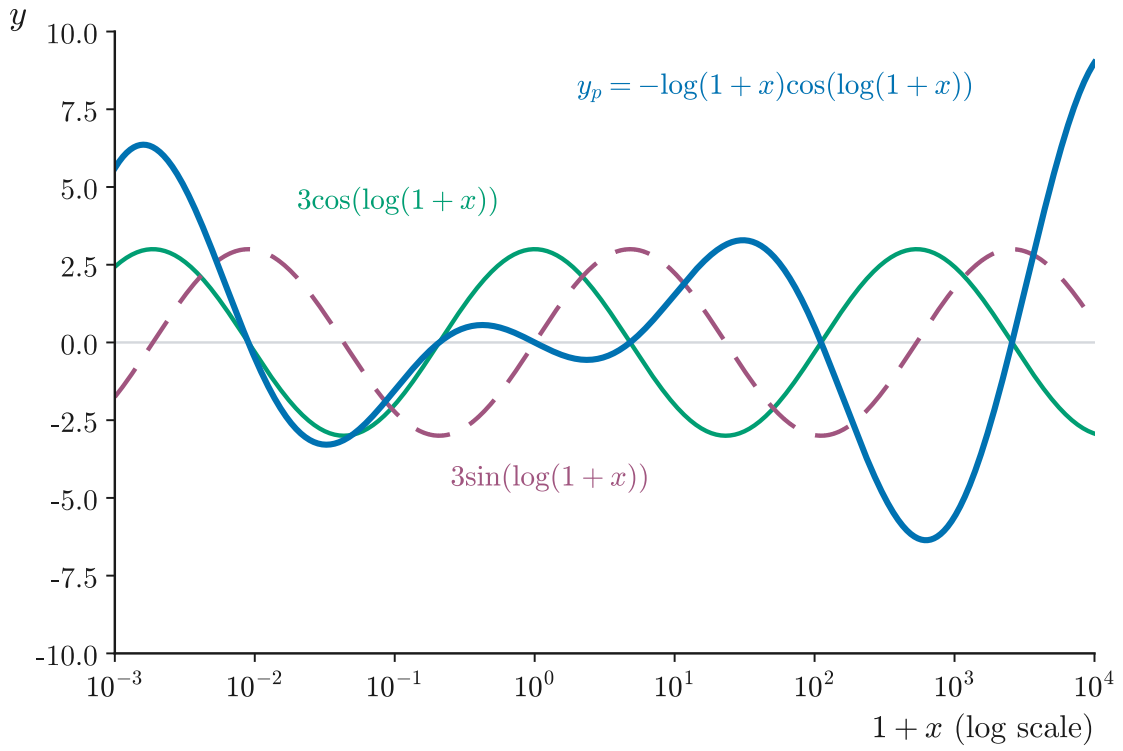


Figure G2.2. Complex roots: $\cos(\log(1+x))$ oscillates ever more slowly. ESA ★★

ADDED FOR CLARITY CW U2 P29 · ESA Dec-23 3b

How to read it. CW P29 and ESA Dec-23 3b have AE roots $\pm i$, so the CF is $c_1\cos(\log(1+x)) + c_2\sin(\log(1+x))$. The horizontal axis is $1+x$ on a log scale, from 10^{-3} to 10^{4}: there the CF terms (green, purple dashed; drawn $\times 3$) are ordinary sinusoids in $z = \log(1+x)$, one period per factor $e^{2\pi} \approx 535$ in $1+x$ — in x the zeros crowd towards $x = -1$ and spread out for large x. Blue: the CW P29 PI $-\log(1+x)\cos(\log(1+x))$, whose amplitude grows like $|\log(1+x)|$ (resonance in z).

3 Legendre problems: the slide example, CW P29–P30 and five ESA questions

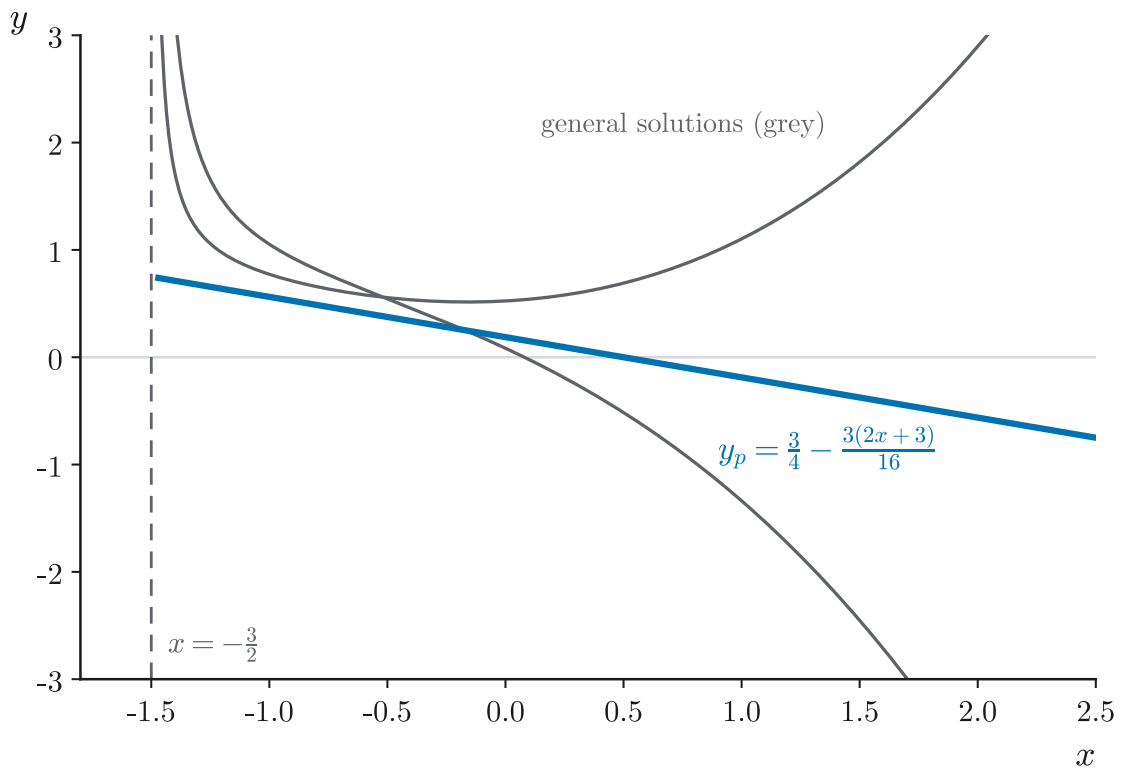


Figure G3.1. Slide example: $y_p = \frac{3}{4} - \frac{3(2x+3)}{16}$ and general solutions. ESA ★★★

ADDED FOR CLARITY S011L·10

How to read it. For $x > -\frac{3}{2}$. Blue: the PI, a straight line $\frac{3}{16} - \frac{3x}{8}$ (a polynomial right-hand side $6x$ gave a polynomial PI). Grey: general solutions with $(c_1, c_2) = (0.01, 0.2)$ and $(-0.01, 0.5)$; the $\frac{c_2}{2x+3}$ term blows up at the asymptote $x = -\frac{3}{2}$ and the $c_1(2x+3)^3$ term dominates for large x.

4 Cauchy problems: lecture examples, CW P31–P33 and ESA Jan-24 s3 3d

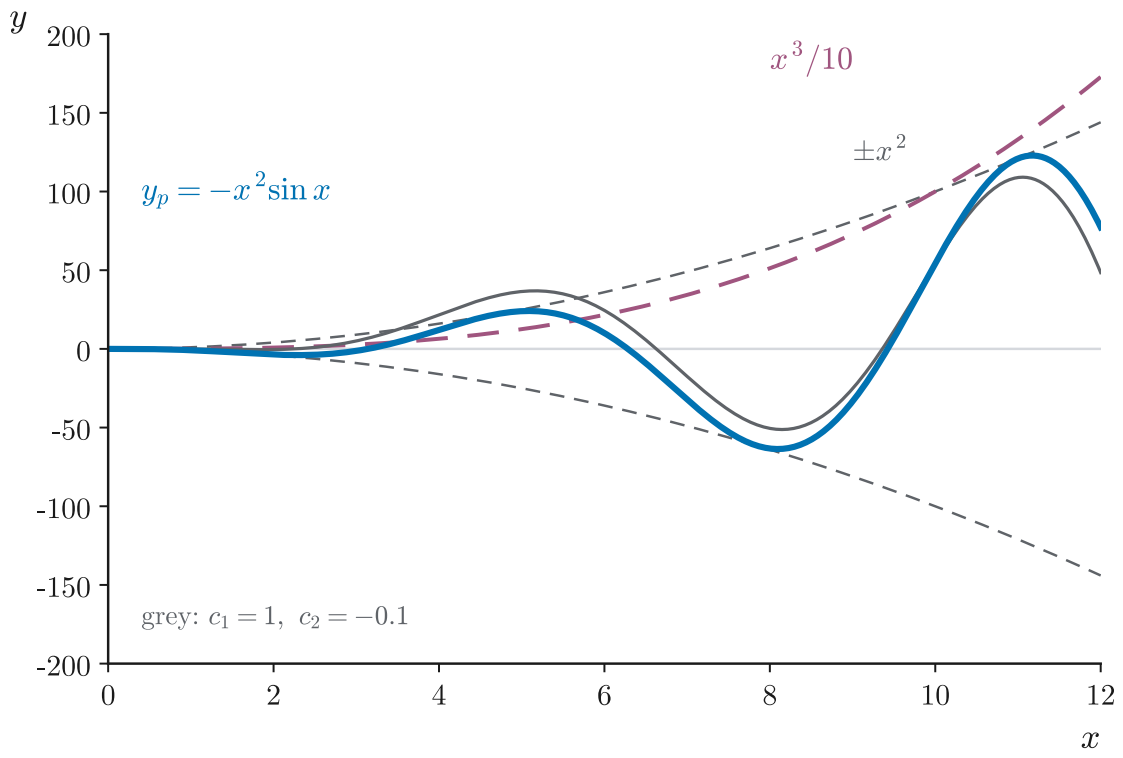


Figure G4.1. ESA Jan-24 s3 3d: $y_p = -x^2 \sin x$ and the CF terms x^2, x^3. ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s3 3d

How to read it. For $x > 0$. Blue: $y_p = -x^2 \sin x$, oscillating between the grey envelopes $\pm x^2$. The upper envelope is the CF term x^2 itself; purple dashed: the CF term $\frac{x^3}{10}$ (scaled). Grey solid: the general solution with $c_1 = 1$, $c_2 = -0.1$.